

Do more with your camera

10 photography hacks from Lifehacker include creating an instant green screen, macro lens and many more: <http://lifehack.kr/digitcam>

We want “HTTPS now”

The Electronic Frontier Foundation together with Access, a digital freedom activist group have started the campaign “HTTPS now” to spread awareness about internet security

up to 15 grand on a camera, so we're not going to attempt pseudo-hogwash of any kind. You know what you want (hopefully), and, even more hopefully, know what you can expect. At this price point you can pick up a super compact camera, or a slightly bulkier one with a longer zoom, a class of camera we have fondly nicknamed “super zooms”. These offer a zoom of between 8x and 12x. A camera with a longer zoom sounds more versatile, but the fact is that you really need to use the zoom before it is actually used. If all you want is to capture photos of family and such, you'll rarely need a zoom greater than 5x. However, this doesn't mean a 12x zoom is bad, as we said, such cameras are more versatile, but they might give up some image quality to a similar sized and similar priced camera, with a smaller optical zoom. Cameras with greater zooms also tend to be bulkier – there's only so much room in which to compress quality optics, and there's a reason why some of Nikon and Canon pro-grade lenses weigh as much as 7 such compact cameras.

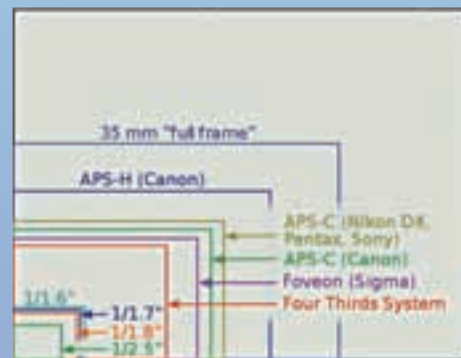
Features

Four cameras – and all reasonably well built specimens. We'll start with the costliest of the bunch, the Nikon Coolpix S8100 – a well-built, sleek looking camera. Made of metal, this S8100 feels very dense in hand and offers a great zoom of 10x, while retaining a slightly slimmer body with a smaller lens bulge than Canons SX130 IS that nonetheless has a slightly longer zoom at 12x. One thing we really loved about the S8100 was its LCD – crisp, bright and in a word – great. The SX130 IS has a dismal LCD particularly when in Live View mode – poor colour, and there is visible ghosting and noise. The Fujifilm T200 was the

SENSORS AND SENSOR SIZES – WHY DO I CARE?

Unlike a film camera, where photosensitive chemicals are exposed to light, any digital camera features an analog device called a sensor. A sensor functionally receives an optical image from a lens, and converts it into an electric signal. Each sensor has millions of photo-diodes. Once exposed to light, these sensors are essentially “charged” and this charge value has to be read for each pixel, and this analog reading is converted to a digital value by other circuitry. This circuitry could be near the photo-diode (CMOS), or off the photo-sensitive area (CCD).

The two most common types of sensors used are CMOS and CCD. These differ in terms of where the charge is read and converted, the size of chip, the cost of manufacture and battery life. In terms of quality, there are about equal these days, though most dSLRs come with CMOS sensors, this is done to keep costs down and increase the number of shots on a single battery charge. More importantly, for a photographer, or even a casual clicker, it is important to know that sensors come in different sizes, and like most things, the larger the better.



Medium format Kodak KAF 3900 sensor

For a sensor to capture more light, you must increase the gain on the sensor, allowing it to amplify the given light. Any amplification of a signal beyond viable limits deteriorates the signal, and in case of camera sensors, this translates to noise. Therefore, a larger surface area given to each photodiode works wonders for their ability to “gather” light, the only problem being the sharply increasing cost of manufacturing a larger chip. This is why manufacturing a full frame sensor of say, a Canon EOS 5D Mark 2 would cost more than manufacturing an entire compact camera, especially if you take into account the amount spent on researching these larger, more complicated sensors.

For example, the sensor on a compact camera measures approximately 6 x 4.5mm giving it a surface area of 27mm², while for a full frame camera this is 36 x 24mm, equalling 864mm². When looking at this from a megapixel perspective, given a fixed megapixel rating, a sensor with a larger surface area would capture more light, requiring less signal amplification to produce the same amount of light as a smaller sensor and thereby producing less noise. Larger sensors have other benefits as well but those are beyond the purview of this test. Below is a look at different common sensor sizes.

other one that stood out – a really slim profile concealing a 10x zoom where we'd figured (given its size) would be closer to 5x. The plastic stick-on in front that is supposed to double as a grip is a pain though – it is anything but grippy, and really Fujifilm could have omitted it to no usability or appearance loss. What we didn't like about the Fujifilm T200 was the actual action of zooming that isn't consistent – it's slow till 2.5x or so then speeds up to 8.6x. From 8.6x to 10x there is essentially a single step with no gradation in between, so you cannot zoom to 9x for example.

The Canon IXUS 130 was the baby zoom in this group offering

a mere 4x optical zoom, while also managing to be the most compact camera in this category and also matching most of the others in terms of build, although if we had to give out a trophy for the best built clicker, it'd have to be the Nikon Coolpix S8100, which also features (interestingly) a backside illumination CMOS sensor – a new technology that is supposed to improve performance in lower light or at higher ISOs, depending on how you look at it.

Canon cameras have a wonderful zoom – it is slower or fast depending on finger pressure applied to the zoom ring – we're coining a phrase for this feature and calling it “variable pressure

zoom”, feel free to come up with something better, and let us know about it. Regardless of the name, it works well, you can quickly zoom, or slowly zoom into anything – very useful for photography and even more useful when shooting videos especially if you want to give a slow pan and zoom effect.

Performance

Although the Coolpix S8100 didn't live up to the high ISO expectancies we had, it did a good job keeping the competition at bay in the high ISO tests. Curiously, at ISO 160, its base ISO, we found the image to be a little flat, lacking the sort of specular highlights and